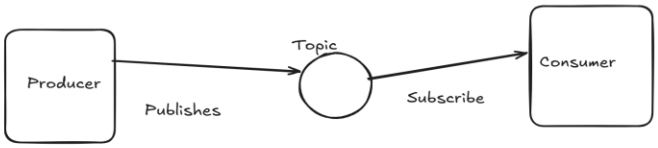
KAFKA

Kafka is Distributed, High Throuput, Realtime Streaming Platform . It is Used to Transport the Large Volumes of data In Real Time Between Two Systems

Kafka



Producer

It is Application that Generates or sends the data to the Topic

Doe

Topic

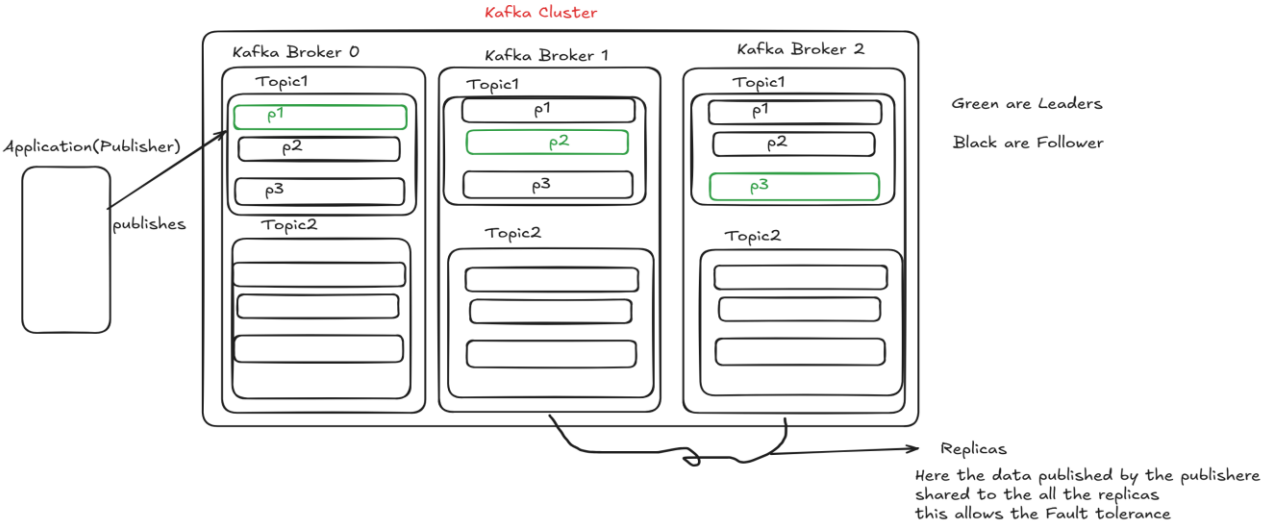
It is a place where the data is Stored in Kafka Broker

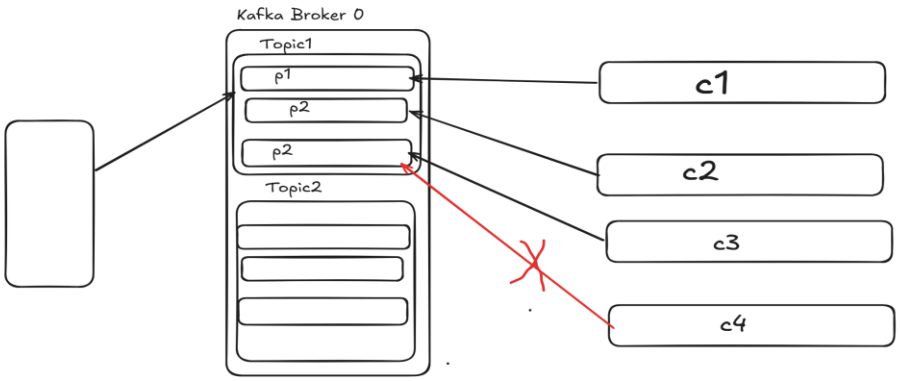
It is Divided into partition To Enhance Performance

Consumer

It is an Application that consumes the data from the Topics

Kafka Architecture





Two consumer cannot consumes the data from same partition
which makes the Redundancy so this not allowed

The consumers operate as Consumer group for load sharing to avoid missing data

Note

The consumer from same group cannot consumes data from same partition consumed by another consumer in same group

But from different group can subscribe the data

Zookeeper

It contains the Entire information about the
Kafka Cluster

No of Brokere

No of Topics

No of partitions in Topics

Who is leader and who are the follwers

If one Kafka Broker got down this zookeeper decised who will be the next Leader
to receive the data from publisher

Leader

It is partition or the Topic in all replicas where the data is received directly from the Publisher

Follower

It contains the replica data of the leader

How to Install Kafka In Windows :

1. Download the kafka From Apache Kafka Website
2. Extract and copy the server and zookeeper file from config to the windows folder to perform operations or don't move

Open cmd as Administration mode for windows folder in kafka

1.Start the Zookeeper First

1. if You copy files into windows execute

```
zookeeper-server-start.bat zookeeper.properties
```

2.start the Server next(another cmd terminal)

```
kafka-server-start server.properties
```

Now Create the two application

1. Producer
2. Consumer

In Producer

Create the kafkaTemplate and send the message and Topic Name

Use Below Properties

```
spring.kafka.producer.bootstrap-servers=localhost:9092
spring.kafka.producer.keyserializer=org.apache.kafka.common.serialization.StringSerializer
spring.kafka.producer.value-serializer=org.apache.kafka.common.serialization.StringSerializer
```

In Consumer

Create KafkaListener and set the Topic and Group Id

Use Below Properties

```
spring.kafka.producer.bootstrap-servers=localhost:9092
spring.kafka.consumer.key-deserializer=org.apache.kafka.common.serialization.StringDeserializer
spring.kafka.consumer.value-deserializer=org.apache.kafka.common.serialization.StringDeserializer
spring.kafka.consumer.group-id=<Group Name>
```